

Datasheet - Smartphone High-Voltage Fast-Charge Cell V27_combo

Doc No: VBF-T6R3-DSH-01	Case: t6_r3	Rev: 01	Date: 2026-08-26
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Electrical

Item	Value	Condition	Source
Nominal capacity	5.0569 Ah	1C discharge, 4.7 -> 2.5 V, DFN	r10_V27_combo_1c.json capacity_ah
Rated energy	21.1125 Wh	discharge integral	r10_V27_combo_energy.json energy_wh
Voltage plateau (midpoint)	4.1551 V	contract metric (calc-energy)	r10_V27_combo_energy.json midpoint_voltage_v
Upper / lower cut-off	4.7 V / 2.5 V	LNMO window	LNMO.json
Initial cell voltage (model initial state)	4.1995 V	cell born nearly full	r10_V27_combo_1c.json voltage_v[0]
DC resistance (DCR)	4.986 mOhm	calc-energy	r10_V27_combo_energy.json dcr_ohm

Energy and power density

Item	Value	Formula / basis	Source
Volumetric energy density	983.61 Wh/L	energy / stack volume (21.4643 mL)	r10_V27_combo_energy.json energy_density_wh_l
Gravimetric energy density	442.80 Wh/kg	energy / dry mass (47.6800 g)	r10_V27_combo_energy.json energy_density_wh_kg
Power density	18544.4 W/kg	calc-energy	r10_V27_combo_energy.json power_density_w_kg
Cell thickness	209 um	stack sum	r10_V27_combo_energy.json thickness_m
Dry / wet mass	47.6800 g / 54.2105 g	wet = dry + electrolyte pore fill	calculated from layers

Fast charge (4C, 18 A) and thermal

Item	Value	Note	Source
Max cell temperature	318.922 K = 45.77 C	limit 50 C (323.15 K); margin 4.23 K	r10_V27_combo_4c.json T_max_K
Min anode potential	+0.1050 V	>= 0 V required; no lithium plating	r10_V27_combo_4c.json anode_potential_v
4C CC phase duration	~16.2 ms	cell starts at 4.1995 V -> 4.7 V cut-off reached almost immediately; charge adds 8.09e-05 Ah (protocol artifact, disclosed - see DVPR row 8)	derived: Q4C/18A
1C max temperature	299.034 K	reference	r10_V27_combo_1c.json T_max_K

Cycle life / durability

Item	Value	Note	Source
Anode SEI thickness after 100 cycles	313.7 nm	limit 500 nm; margin 186.3 nm	r10_V27_combo_aging.json sei_thickness_nm_end
Initial SEI thickness	5 nm	Chen2020 base	Chen2020.py:240
Capacity trajectory	cycle 1: -0.618 Ah -> cycle 100: 2.732 Ah	SEI-model lithium-inventory climb artifact; NOT a real capacity gain; durable metric is SEI thickness	r10_V27_combo_aging.json capacity_ah_per_cycle

Composition (dry cell, contract basis)

Component	Mass	Share
LNMO spinel active (positive)	17.3088 g	36.3 %
Graphite active (negative)	13.2161 g	27.7 %
Carbon black conductive additive	0.7780 g	1.6 %
PVDF binder	0.3606 g	0.8 %
SBR/CMC binder	0.2782 g	0.6 %
Separator (polyolefin, 12 um)	0.2593 g	0.5 %
Aluminium foil (16 um)	4.4366 g	9.3 %
Copper foil (12 um)	11.0423 g	23.2 %

Electrolyte 6.5305 g (5.44 mL pore volume at 1.2 g/mL) is excluded from the contract mass and energy-density basis (electrolyte_included=False in energy output).

Notes

- All performance values are DFN simulation outputs (model_used=DFN). No physical cell exists; no true DFT/MD was run (real_compute=false, endorse skipped honestly).
- Nominal reference capacity of the base set is 4.5 Ah [LNMO.json]; the simulated 1C discharge yields 5.0569 Ah under the 2.5-4.7 V window.
- N/P (layer capacity ratio) = 1.5787, first-principles formula; early probe labels used a thickness-scaled shorthand.